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Project ID: PID-4.7

Magnetic Lock Sensor

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Table of contents

01

Abstract

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Abstract

The Magnetic Lock Sensor (MLS) project aims to enhance the security of personal belongings during travel, particularly in **crowded public transport scenarios**. The system includes two modules:

- the Tracker Module: This module is equipped with an Arduino UNO, Hall-Effect Sensor and HC-05 Bluetooth Module, detects the **Presence or Absence of a Magnetic field from the Auxiliary Magnet to determine the lock status.**

- the Indicator Module: This module also features an Arduino UNO, HC-05 Bluetooth Module, Piezo Buzzer and RGB LED. It provides **Visual and Auditory-alerts to indicate the status of the lock.**

Both modules are powered by a battery pack and communicate wirelessly with a mobile device. MLS has new innovative & enhanced security features, improved user experience, a tracking feature, compact design, and battery backup.

These features aim to provide users with a reliable and efficient solution for securing their belongings and receiving immediate alerts in case of unauthorized access or potential theft.

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02

Objective

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Objective

1. Enhancing Luggage and Personal Belongings Safety:

The Magnetic Lock Sensor (MLS) project aims to ensure the safety and security of luggage and personal belongings during travel, especially in crowded public transport scenarios, by providing immediate alerts in case of unauthorized access or potential theft.

2. Efficient Tracking in Case of Theft or Loss :

MLS includes a tracking feature that activates if the luggage is transported to a different location. It sends location updates after short delays, enabling users to quickly locate their belongings in case of theft or loss. This tracking feature enhances the overall security of belongings and provides users with peace of mind.

3. Cost-efficient and Effective Anti-theft Security Device:

MLS is designed to be cost-effective while providing effective anti-theft security. It includes features such as a multi-color RGB LED and a positive buzzer for visual and auditory alerts, making it a reliable and efficient solution for securing belongings without the need for expensive security systems.

03

Introduction

Introduction

Traveling in crowded public transport, such as buses or trains, often presents the challenge of keeping luggage and personal belongings secure. Items like backpacks, handbags, and wallets are prone to being misplaced, forgotten, stolen, or accessed without authorization.

This problem is particularly concerning for daily commuters and even individuals with disabilities, such as those who are blind or deaf.

To address these issues, we introduce the **Magnetic Lock Sensor (MLS)** as a solution. MLS enhances the safety of personal belongings by providing immediate alerts in case of unauthorized access or potential theft. The device employs a combination of a :

- **Hall effect sensor, RGB LED, and piezo buzzer**
- **To offer visual and auditory alerts.**
- **Additionally, MLS includes a tracking feature**
- Sending **real-time location updates**, ensuring the user can quickly recover their belongings if they are Stolen/misplaced.

This cost-effective and efficient anti-theft device aims to provide peace of mind and security for travelers.

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04

Social Impact

Social Impact

The Social Impact of the Magnetic Lock Sensor (MLS) can be profound, especially in the context of public transit systems. Here are some key points:

1. **Enhanced Safety and Security:** MLS provides individuals with peace of mind by ensuring the safety of their belongings.

2**. Reduced Stress and Anxiety**: Knowing that their belongings are secure, commuters can travel without constantly worrying about their luggage. This can significantly reduce stress and anxiety, especially for frequent travelers.

3**. Financial Savings**: The minimal cost of MLS compared to the potential loss of valuable belongings due to theft or misplacement represents a significant financial saving for individuals.

4**. Efficiency in Travel:** By providing immediate alerts and tracking options, MLS helps travelers react promptly in case of a security breach. This can lead to a more efficient travel experience.

5. **Empowerment of Vulnerable Groups:** MLS can be particularly beneficial for vulnerable groups such as the elderly, disabled, or individuals with sensory impairments who may find it challenging to keep track of their belongings in crowded environments.

6. **Community Safety**: By deterring theft and unauthorized access to luggage, MLS contributes to the overall safety and security of the community using public transit systems.

7. **Reduction in Crime:** The presence of MLS in luggage can deter potential thieves, leading to a decrease in theft-related crimes in public transport settings.

Overall, the social impact of MLS extends beyond just the individual user to create a safer and more secure environment for all commuters, enhancing the overall travel experience and promoting a sense of community well-being.

05

Solution Approach

Solution Approach

1. **Innovative Approach**: A novel approach is taken by using magnetic fields to detect a locking mechanism, ensuring reliable security.

2. **Modular Design**: MLS consists of separate modules for tracking and indicating, enhancing flexibility and usability for users.

3. **Problem-Centric Development**: MLS is developed to address the issues of theft, misplaced luggage, and unauthorized access in public transit.

4. **Robust Testing**: MLS undergoes rigorous testing to ensure its effectiveness in real-world scenarios, providing reliable security solutions.

5. **Tailored Deployment**: MLS is deployed in public transit systems to offer enhanced security, addressing travelers' concerns about luggage safety.

06

Model Demonstration

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Model Demonstration

06

Cost Involvement

Cost Involvement

Component	Approximate Price (Rs)
Hall Effect Sensor	50
Microcontroller (Arduino)	300
Bluetooth Module	150
Battery	100
Buzzer	50
LED	20
Switch	20
Magnet	30
Miscellaneous (wires, PCB)	100
Enclosure	200
Total	1020

06

References

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Introduction

Bhattacharya, S., & Sengupta, S. (2020). Development of an IoT-based Smart Luggage Security System. In 2020 IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES) (pp. 1-5). IEEE.

Ravi, R., Gnanasekaran, M., & Muralidharan, S. (2017). A Review on Smart Luggage Security System. International Journal of Innovative Research in Computer and Communication Engineering, 5(5), 8223-8228.

Akinbode, R., Adefila, K., & Adegbola, A. A. (2020). Design and implementation of a smart luggage security system using IoT. Journal of King Saud University-Computer and Information Sciences.

Sharma, V., Singh, N., & Kumar, S. (2021). Design and Development of Smart Luggage Security System using IoT. In 2021 IEEE International Conference on Innovative Research and Development (ICIRD) (pp. 1-6). IEEE.

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